Structural reliability updating through proof load testing

A Bayesian methodology applied to reinforced concrete road bridges and viaducts

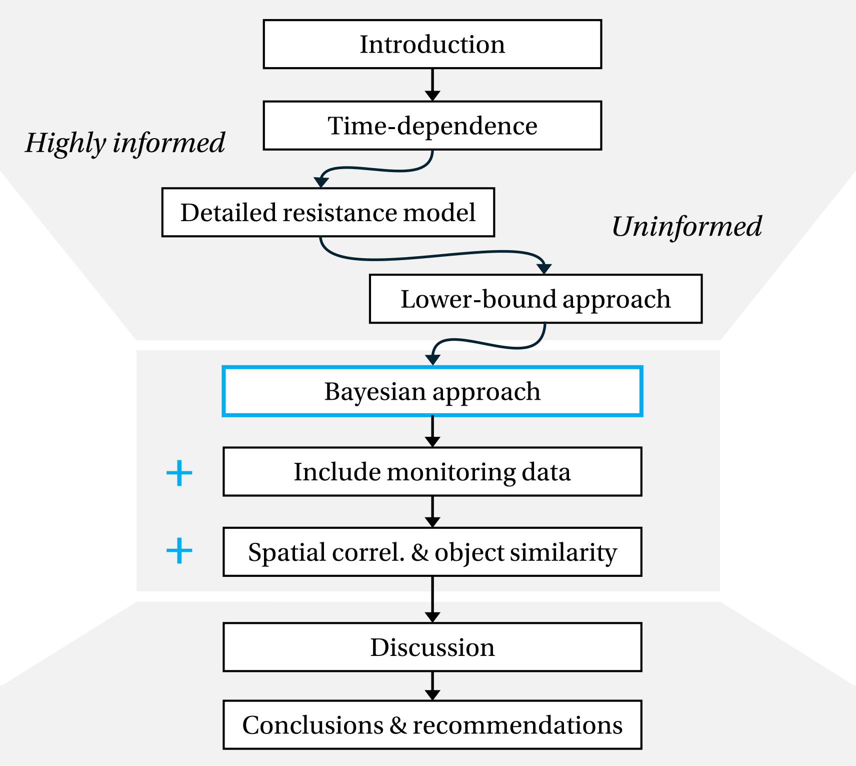
Rein de Vries

Introduction

- Ageing infrastructure raises concerns about safety and reliability → *accurate* structural assessment methods are required
- Pilot load tests performed, but challenges remain in its probabilistic substantiation

Diagnostic ———— $\boxed{\text{Proof load}}$ ———— Collapse

*How can **proof** load testing establish the structural **reliability** of reinforced concrete bridges and viaducts?*


Introduction
Time-dependence
Highly informed
Detailed resistance model
Uninformed
Lower-bound approach
Bayesian approach
+
Include monitoring data
+
Spatial correl. & object similarity
Discussion
Conclusions & recommendations

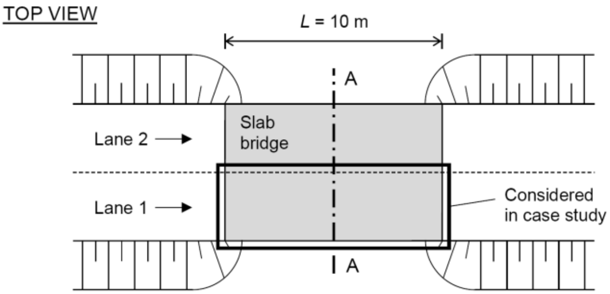
Time-dependent reliability assessment

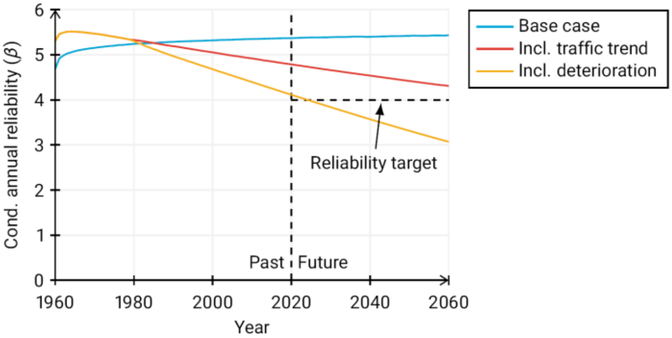
De Vries, R., Steenbergen, R. D. J. M. & Maljaars, J. (2023). Annual reliability requirements for bridges and viaducts. Heron, 68(2), 91–122. https://heronjournal.nl/68-2/1.html

De Vries, R., Lantsoght, E. O. L., Steenbergen, R. D. J. M. & Fennis, S. A. A. M. (2022). Reliability assessment of existing reinforced concrete bridges and viaducts through proof load testing. Proceedings of the 11th International Conference on Bridge Maintenance, Safety and Management (IABMAS), Barcelona, Spain, 467–475. https://doi.org/10.1201/9781003322641-54

Time-dependent reliability assessment

- Structural reliability is *inherently* time-dependent due to variable load, possible load trends and deterioration
- We can pinpoint the exact moment a bridge's reliability is insufficient

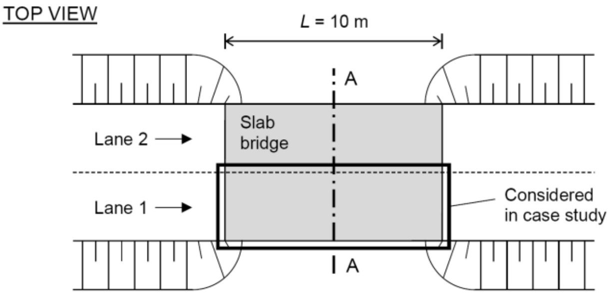


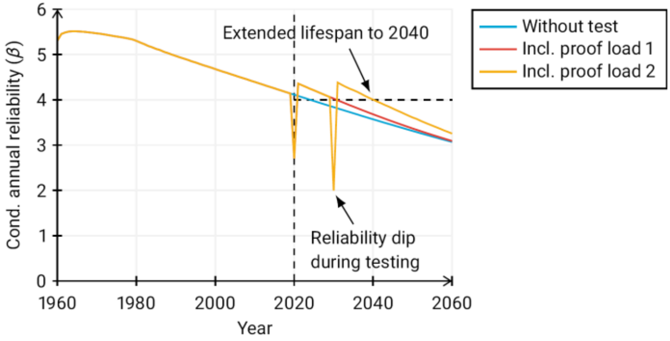


Time-dependent reliability assessment

- Verify structural reliability using proof load testing
- But, what *target load*?

- Temporary reliability drop during testing, but increased reliability after success
- Strategic planning of test timing and target loads





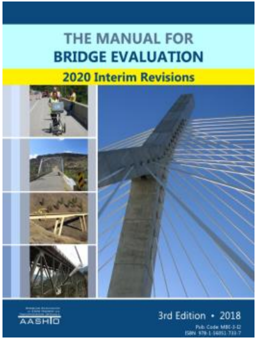
Lower-bound approach based on the load effect

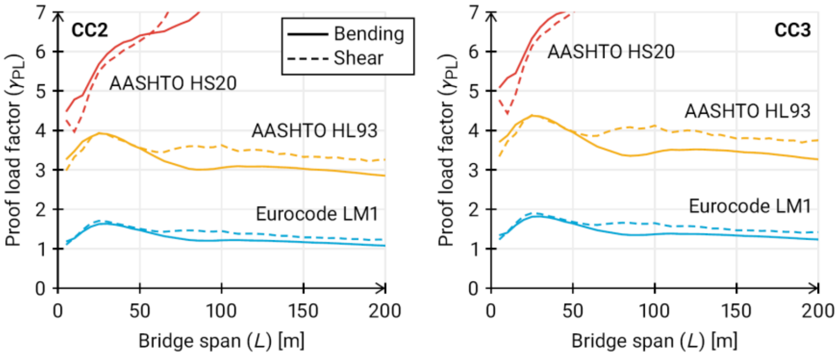
De Vries, R., Lantsoght, E. O. L., Steenbergen, R. D. J. M. & Naaktgeboren, M. (2023). Proof load testing method by AASHTO and suggestions for improvement. *Proceedings of the 14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP), Dublin, Ireland.* http://hdl.handle.net/2262/103200

De Vries, R., Lantsoght, E. O. L., Steenbergen, R. D. J. M. & Naaktgeboren, M. (2023). Proof load testing method by the American Association of State Highway and Transportation Officials and suggestions for improvement. *Transportation Research Record, 2677*(11), 245–257. https://doi.org/10.1177/03611981231165026

Lower-bound approach

- As suggested in American MBE
- Characteristic traffic load multiplied by a target proof load factor
- Proof load factors derived using Dutch WIM data
- Practical and straightforward, but conceptual challenges remain





Bayesian approach to address varying knowledge levels

De Vries, R., Lantsoght, E.O. L., Steenbergen, R.D. J. M. & Fennis, S. A. A. M. (2023). Time-dependent reliability assessment of existing concrete bridges with varying knowledge levels by proof load testing. *Structure and Infrastructure Engineering: IABMAS 2022 Special Issue, 20*(7-8), 1053–1067. https://doi.org/10.1080/15732479.2023.2280712

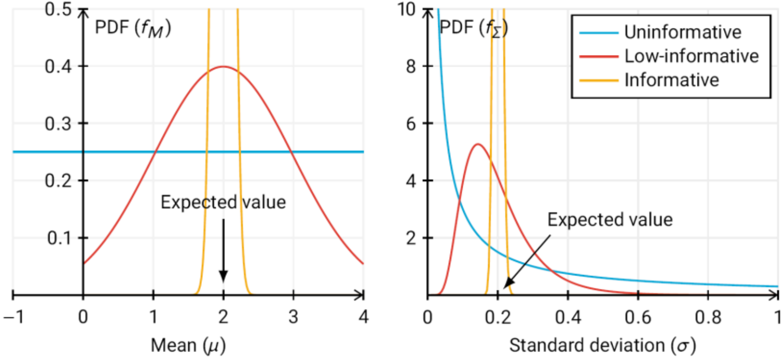
Bayesian approach

- Limited resistance information should still be incorporated
- Bayesian methodology integrates all available (uncertain) information

Uninformative ——— Low-informative ——— Informative

$$p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)\, p(\theta)$$

$$\theta = (\mu, \sigma)$$

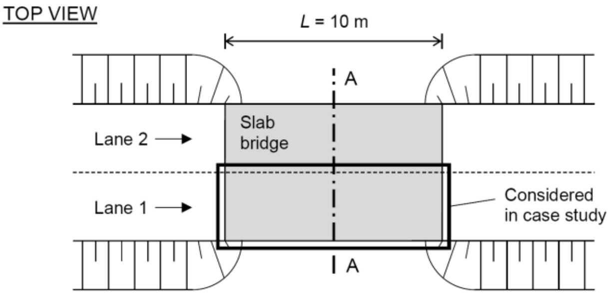


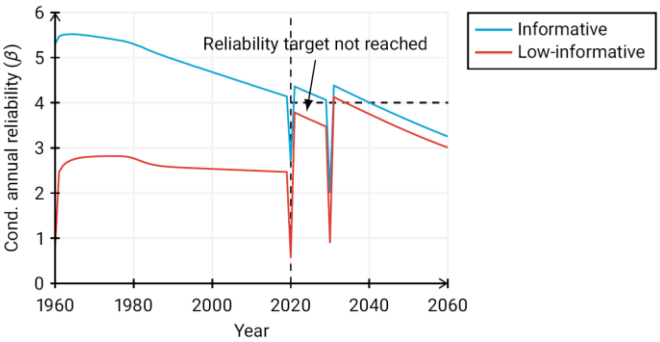
Bayesian approach

- Case study repeated
- No resistance model → only a distribution function

- Desired reliability not reached with the same target loads
- Higher (predicted) failure probability during testing

- Compensate uncertainty with higher test loads





Incorporating monitoring data into the assessment

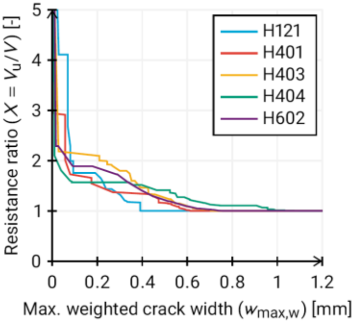
De Vries, R., Lantsoght, E.O. L., Steenbergen, R.D. J. M., Hendriks, M. A. N. & Naaktgeboren, M. (2024). Structural reliability updating using monitoring data from in-situ load testing and laboratory test results. *Proceedings of the 13th International Conference on Bridge Maintenance, Safety and Management (IABMAS), Copenhagen, Denmark*, 409–417. https://doi.org/10.1201/9781003483755-44

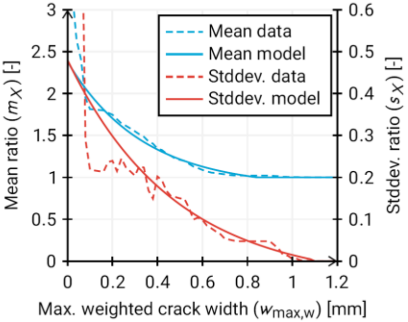
De Vries, R., Lantsoght, E.O. L., Steenbergen, R.D. J. M., Hendriks, M. A. N. & Naaktgeboren, M. (2025). Structural reliability updating on the basis of proof load testing and monitoring data. *Engineering Structures*, *330*, 119863. https://doi.org/10.1016/j.engstruct.2025.119863

Incorporating monitoring data

- Bayesian updating combines in-situ monitoring and laboratory data after each load increment
- Use lab data (or simulations) to establish a relation between *indicator* and *resistance ratio*

- Case studies show test load reductions of 20% and 25%
- Reliability-based alternative for stop criteria





Addressing spatial correlation and system reliability

De Vries, R., Lantsoght, E.O. L., Steenbergen, R.D. J.M., Hendriks, M. A. N. & Fennis, S. A. A.M. (2026). Addressing spatial correlation and system reliability in proof load testing of reinforced concrete bridges and viaducts. *Structural Safety, Under review*.

De Vries, R., Steenbergen, R.D. J.M. & Vrouwenvelder, A. C.W.M. (2025). Bayesian structural reliability updating using a population track record. *Reliability Engineering & System Safety, 225*, 110644. https://doi.org/10.1016/j.ress.2024.110644

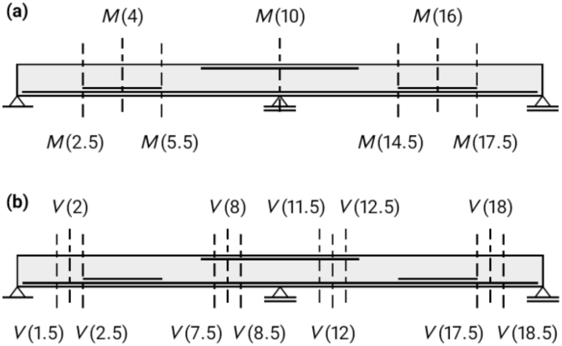
Spatial correlation and system reliability

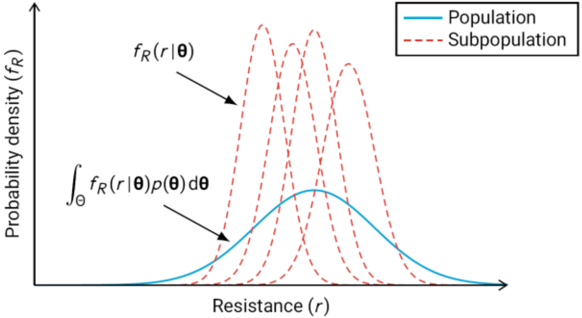
- All components and failure modes are difficult to validate using a few tests
- System-reliability updating framework based on a *hierarchical* Bayesian model

Two applications

1. Transfer factors
2. Optimal testing vehicle layout

Fewer tests ———— Lower load levels





Conclusions

Conclusions

- Combined with probabilistic modelling, proof load testing enables *uniquely accurate* structural assessments
- Bayesian methods can incorporate varying levels of information
- Monitoring data enables real-time reliability updating and rational stop criteria
- Spatial correlation and system reliability allow limited tests to inform untested components

Recommendation for future research

- Improve statistical models of traffic loads (e.g. multi-lane, longer spans)
- Enable simplified tools (e.g. pre-computed tables)
- Expand monitoring techniques (e.g. acoustic emission, long-term sensing, …)

Thank you

TU Delft

TNO innovation for life

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